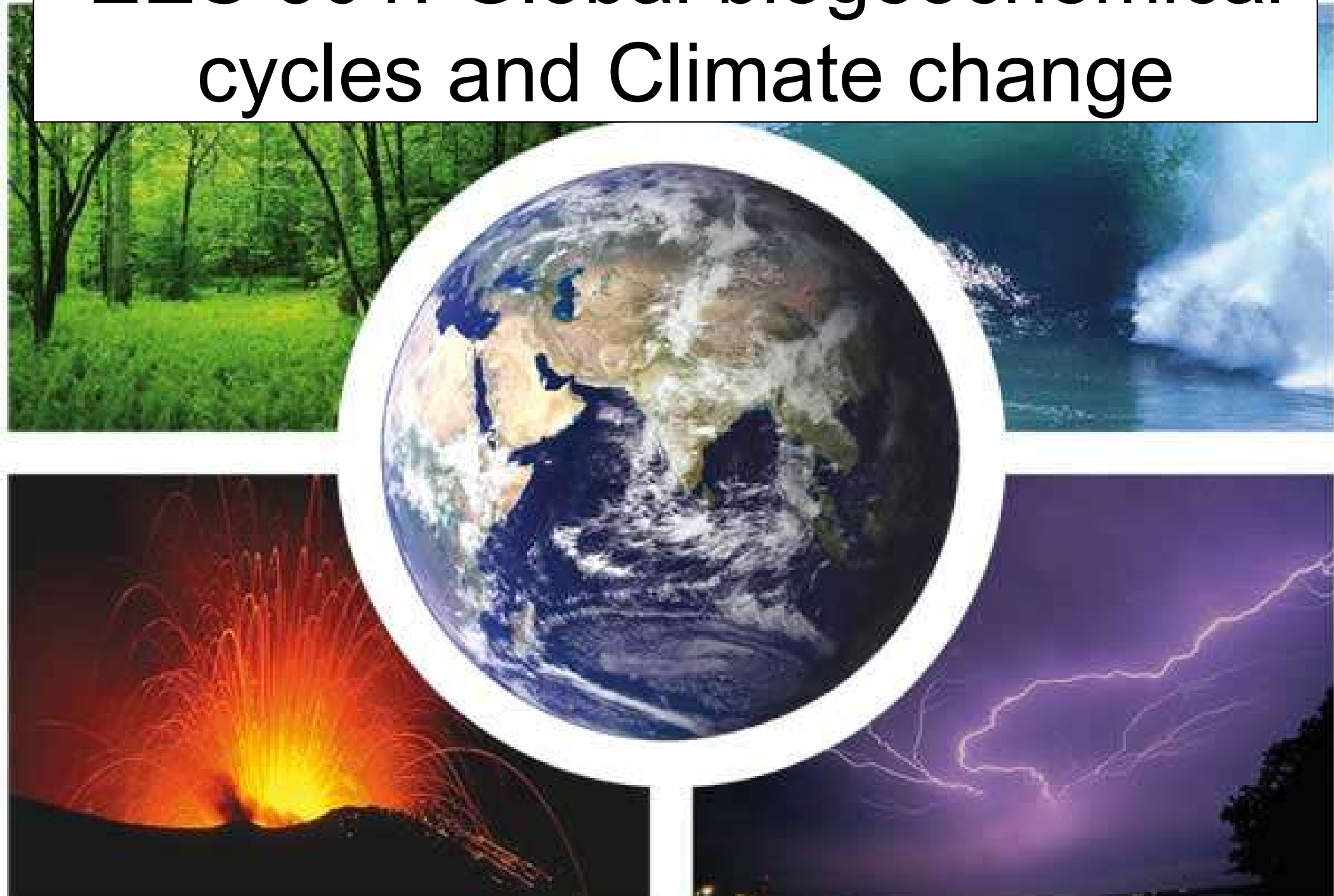


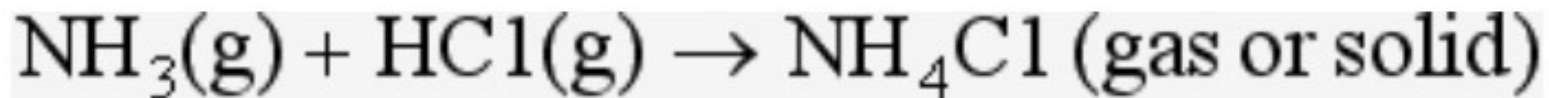
EES 301: Global biogeochemical cycles and Climate change



The acid base and oxidation-reduction balances of the Earth System

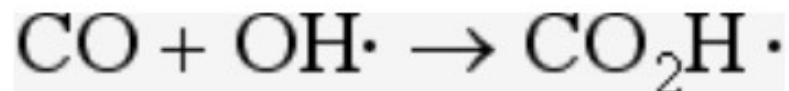
- Most of what we have learned about p_ϵ (or E_h) and pH is directly applicable to liquid phases
- While redox reactions do occur in the gas phase all the time, their rates are typically governed by kinetics rather than thermodynamics.

The acid base reaction:



... is governed by 2nd order kinetics. However in this case how much solid you get on your filter versus how much of the product will be in the gas phase will be governed by thermodynamics. The equilibrium is strongly T & RH dependent.

The redox reaction:



... is governed by pseudo-first order kinetics because CO is very much in excess in relation to the OH radical

The acid base and oxidation-reduction balances of the Earth System

➤ pH and p_ϵ are defined as

$$\begin{aligned} \text{pH} &= -\log_{10}[\text{H}^+] \\ p_\epsilon &= -\log_{10}[\text{e}^-] \end{aligned}$$

- A low pH means a large value of $[\text{H}^+]$ and high acidity
- A low p_ϵ means a large value of $[\text{e}^-]$ and strongly reducing conditions
- Acid base reactions involve the exchange of protons between the reactants
- Oxidation reduction reactions on the other hand involve the exchange of electrons and a change in the oxidation state of the reactants involved

The acid base and oxidation-reduction balances of the Earth System

- These types of reactions happen wherever water is involved. In:
 1. clouds
 2. activated aerosol particles
 3. rainwater & snow
 4. surface runoff
 5. groundwater
 6. rivers and streams
 7. lakes
 8. the ocean

Key acids and bases

Table 16-1

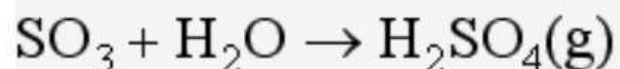
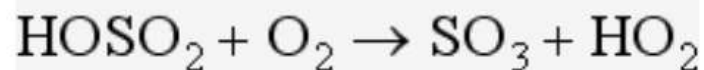
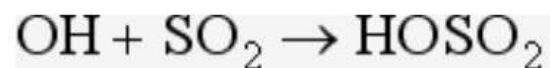
Dominant acids and bases in the Earth system

A.	Acids	
	1.	Carbon based:
		H ₂ CO ₃ (carbonic acid) aqueous phase
		HC ₂ H ₃ O ₂ (acetic acid) aqueous, gas phases
		HCHO ₂ (formic acid) aqueous, gas phases
		H ₂ C ₂ O ₄ (oxalic acid) aerosol particles; solid phase
		RCOOH (many carboxylic acids)
	2.	Nitrogen based:
		HNO ₃ (nitric acid) gas, aqueous, solid phases
		HNO ₂ (nitrous acid) gas, aqueous phases

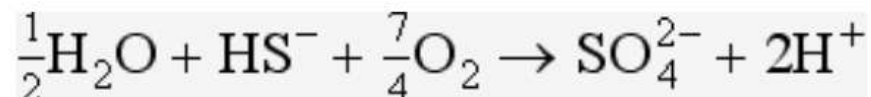
	3.	Sulfur based:
		H ₂ SO ₄ (sulfuric acid) aerosol, aqueous phases
		H ₂ SO ₃ (sulfurous acid) aqueous phase
		CH ₃ SO ₃ H (from oxidation of (CH ₃) ₂ S, methane sulfonic acid) aerosol, aqueous phases
B.	Bases	
	1.	Nitrogen based:
		NH ₃ , RNH ₂ , R ₂ NH, R ₃ N (ammonia and amines) gas and aqueous phases
	2.	Igneous mineral based:
		Al, Mg silicates (weathering reaction in Chapter 8 and 9)
	3.	Sedimentary mineral based (examples):
		CaCO ₃
		Al(OH) ₃
		(Chapter 8 and 9)

Key acids and bases in air and their formation

- Gas phase examples:



- Aqueous phase example:



Let us consider a CO₂ as an acid

$$K_H = 29 \text{ atm mol}^{-1} \text{ L}$$



To calculate M of H₂CO₃^{*} for water in equilibrium with a partial pressure of 300 ppm CO₂ which is the same as 300x10⁻⁶ atm then:

$$\begin{aligned} & 300 \times 10^{-6} \text{ atm} / (29 \text{ atm mol}^{-1} \text{ L}) \\ & = 1 \times 10^{-5} \text{ mol/L} \end{aligned}$$

Next to calculate pH:

If you have to calculate pH then you need the K₁ = 4.5 x 10⁻⁷ mol/L



$$K_1 = 4.5 \times 10^{-7} = [\text{HCO}_3^-][\text{H}^+] / [\text{H}_2\text{CO}_3] = [\text{HCO}_3^-]^2 / 1 \times 10^{-5}$$

$$[\text{HCO}_3^-] = [\text{H}^+] = 2.1 \times 10^{-6} \text{ M/L};$$

$$\text{pH} = -\log_{10}(a_{\text{H}^+}) \approx -\log_{10}([\text{H}^+]/\text{M})$$

$$\text{pH} = 5.6$$

Let us consider a cloud in clean marine air

- In a remote location far from industrial sources e.g. Cape Grim in the southern hemisphere, most CCN are ammonium sulfate with DMS oxidation being the sulfur source. The PM_1 mass loading is typically very less $0.25 \mu\text{g m}^{-3}$
- Because NH_3 gas availability is less the salt will typically be NH_4HSO_4
- A typical aerosol water content would be 0.125 g m^{-3}
- If we want to calculate how many mol of sulfate SO_4^{2-} there would be in the water in mol/L

$$\begin{aligned} [\text{SO}_4^{2-}] &= \frac{2.5 \times 10^{-7} \text{ g / m}^3}{(1.25 \times 10^{-1} \text{ g / m}^3)(96 \text{ g / mol})} \\ &\approx 2 \times 10^{-8} \frac{\text{mol}}{\text{gH}_2\text{O}} \approx 2 \times 10^{-5} \text{ mol / L} \end{aligned}$$

- To estimate the pH consider: $\text{NH}_4\text{HSO}_4 \leftrightarrow \text{NH}_4^+ + \text{H}^+ + \text{SO}_4^{2-}$
Because dissociation constant is large most is there in the dissociated form so

$$[\text{NH}_4^+] = [\text{H}^+] = [\text{SO}_4^{2-}] \approx 2 \times 10^{-5} \text{ mol / L}$$

$$\text{pH} = -\log_{10}(a_{\text{H}^+}) \approx -\log_{10}([\text{H}^+]/M)$$

$$\text{pH} \sim 4.7$$

Firstly this tiny amount of salt causes one order of magnitude (10x) more protons in the solution than 300 ppm CO_2 would, so aerosol acidity is usually not governed by CO_2 but by other acids. However, this is only the pH the aerosol would have if this salt was the only one present.

Let us consider a cloud in clean marine air

➤ In reality, there are many other compounds that could influence pH which we need to consider and that is just when we include the two precursors of the salt $\text{SO}_2(\text{g})$ which gets oxidized to gas phase acid to react with ammonia $\text{NH}_3(\text{g})$ gas.

$$\log[A^-] - \log[HA] = \text{pH} + \log K_a$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$\log K_a = \log[H^+] + \log[A^-] - \log[HA]$$

or

$$\log[A^-] - \log[HA] = \text{pH} + \log K_a$$

Because this is in the simple algebraic form:

$$y = mx + b$$

We can plot all of these equations with pH on the x-axis

Table 16-2

List of input components for the simplest case of the acid-base balance of unpolluted marine clouds. Also shown are the mass conservation statements, chemical equilibrium expressions and constants, and the requirement for charge balance

Equilibrium	Equilibrium constant	Assumed value at 5°C
(1) $\text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{OH}^-$	$K_w = [\text{H}^+][\text{OH}^-]$	$1.82 \times 10^{-15} \text{ mol}^2/\text{L}^2$
(2) $\text{SO}_2(\text{g}) + \text{H}_2\text{O} \leftrightarrow \text{SO}_2 \cdot \text{H}_2\text{O}$	$K_{\text{HS}} = \frac{P_{\text{SO}_2}}{[\text{SO}_2 \cdot \text{H}_2\text{O}]}$	0.379 atm L/mol
(3) $\text{SO}_2 \cdot \text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{HSO}_3^-$	$K_{\text{IS}} = \frac{[\text{H}^+][\text{HSO}_3^-]}{[\text{SO}_2 \cdot \text{H}_2\text{O}]}$	0.0206 mol/L
(4) $\text{HSO}_3^- \leftrightarrow \text{H}^+ + \text{SO}_3^{2-}$	$K_{\text{2S}} = \frac{[\text{H}^+][\text{SO}_3^{2-}]}{[\text{HSO}_3^-]}$	$8.88 \times 10^{-8} \text{ L}$
(5) $\text{NH}_3(\text{g}) + \text{H}_2\text{O} \leftrightarrow \text{NH}_3 \cdot \text{H}_2\text{O}$	$K_{\text{HN}} = \frac{P_{\text{NH}_3}}{[\text{NH}_3 \cdot \text{H}_2\text{O}]}$	$7.11 \times 10^{-3} \text{ atm L/mol}$
(6) $\text{NH}_3 \cdot \text{H}_2\text{O} \leftrightarrow \text{NH}_4^+ + \text{OH}^-$	$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3 \cdot \text{H}_2\text{O}]}$	$1.5 \times 10^{-5} \text{ mol/L}$
(7) $\text{CO}_2(\text{g}) + \text{H}_2\text{O} \leftrightarrow \text{CO}_2 \cdot \text{H}_2\text{O}$	$K_{\text{HC}} = \frac{P_{\text{CO}_2}}{[\text{CO}_2 \cdot \text{H}_2\text{O}]}$	16.6 atm L/mol
(8) $\text{CO}_2 \cdot \text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{HCO}_3^-$	$K_{\text{1C}} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2 \cdot \text{H}_2\text{O}]}$	$2.94 \times 10^{-7} \text{ mol/L}$
(9) $\text{HCO}_3^- \leftrightarrow \text{H}^+ + \text{CO}_3^{2-}$	$K_{\text{2C}} = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{CO}_2 \cdot \text{H}_2\text{O}]}$	$2.74 \times 10^{-11} \text{ mol/L}$
(10) $\Sigma[\text{CO}_2] = C_1 \text{ (mol/m}^3\text{)}$		
(11) $\Sigma[\text{S(IV)}] = C_2 \text{ (mol/m}^3\text{)}$		
(12) $\Sigma[\text{N(-III)}] = C_3 \text{ (mol/m}^3\text{)}$		
(13) $\Sigma[\text{Strong acid anions}] = C_4 \text{ (mol/m}^3\text{)}$		
(14) Liquid water content = (L/m^3)		
(15) $\Sigma[+] = \Sigma[-]$		

Let us consider a cloud in clean marine air

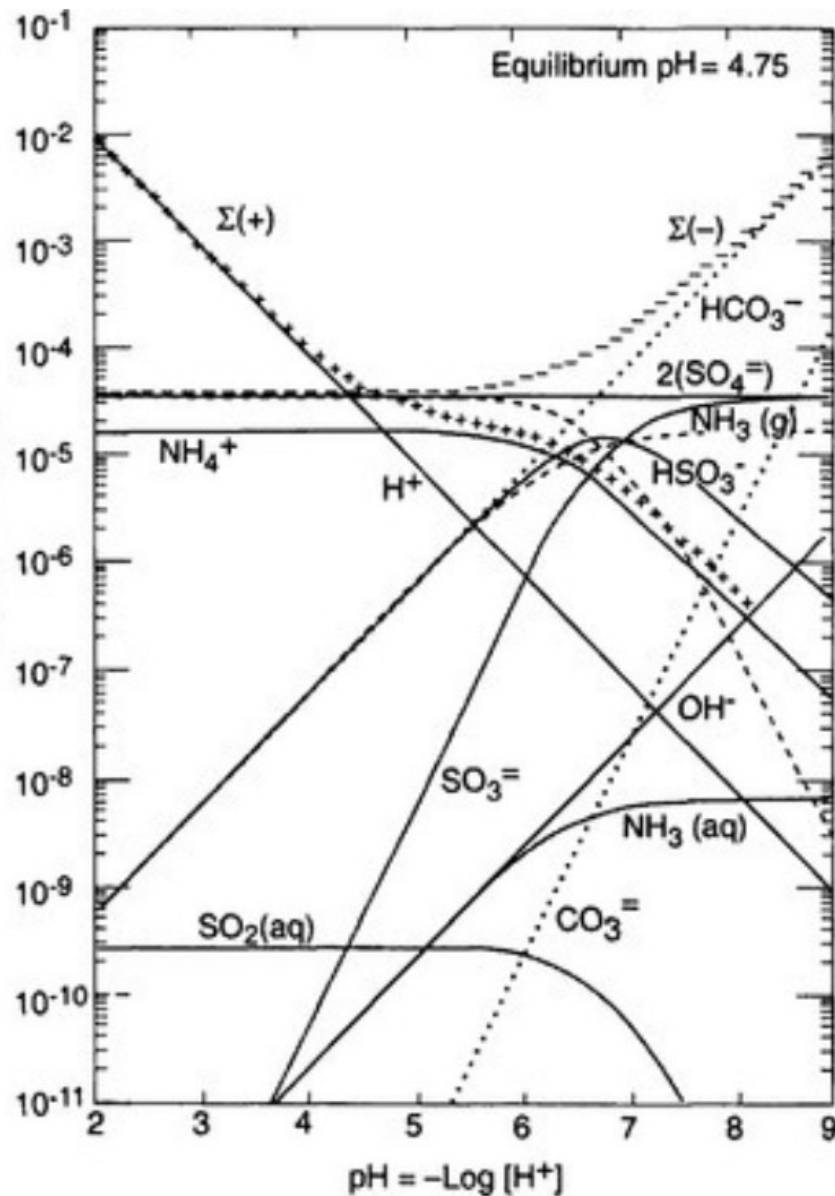
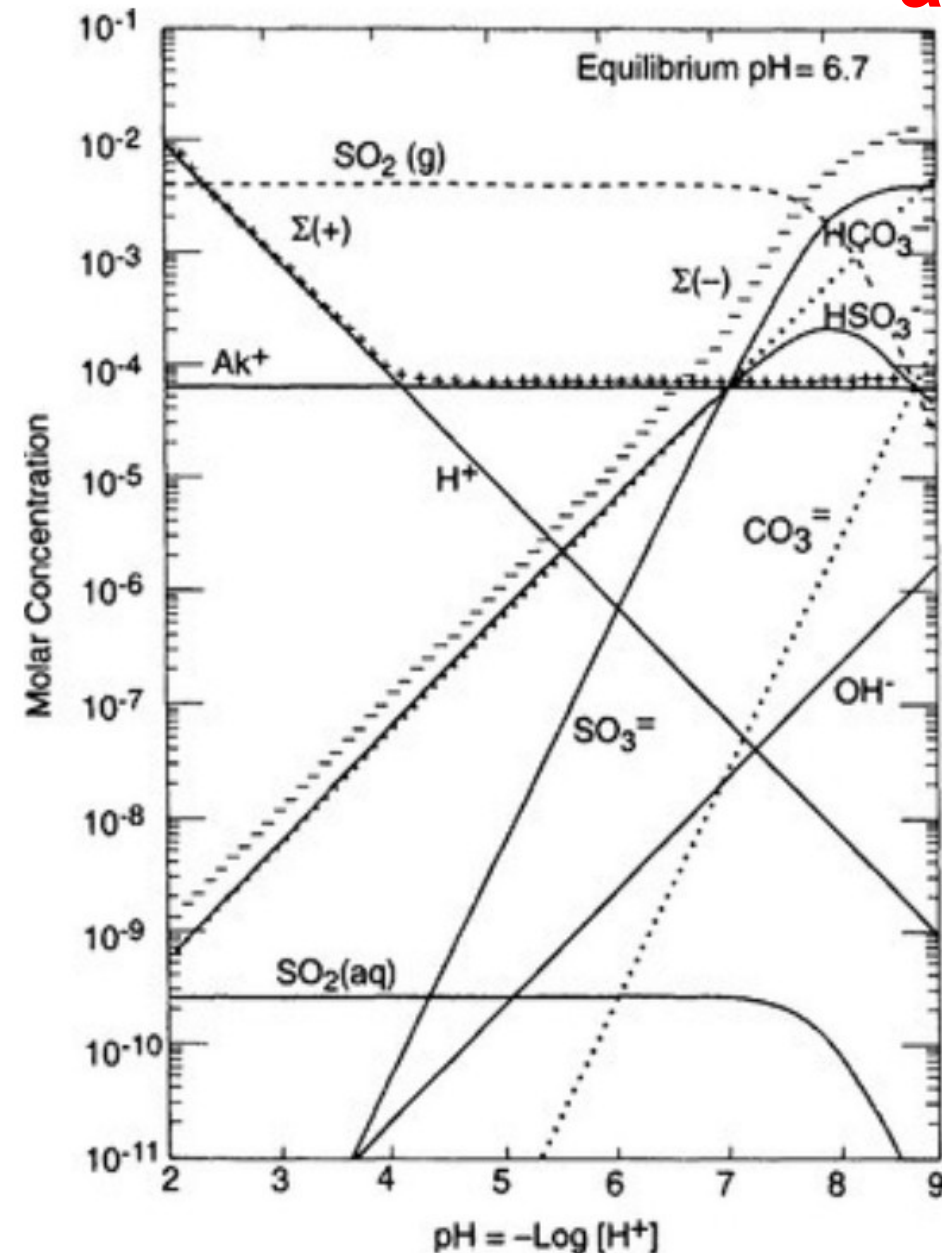


FIG. 16-1 Master-variable diagram of clean marine cloud at a model altitude of 875 m. Equilibrium occurs where $\Sigma[+] = \Sigma[-]$, i.e. charge balance. Input conditions are $0.2 \mu\text{g}/\text{m}^3$ of SO_4^{2-} aerosol, roughly half neutralized by NH_4^+ in the amount $0.039 \mu\text{g}/\text{m}^3$. 0.1 ppb by volume SO_2 and $0.125 \text{ g}/\text{m}^3$ of liquid water, 340 ppm CO_2 , temperature is 278 K . (After [Twohy et al., 1989](#).)

The stronger acid pushes the weaker acid into the gas phase

Equilibrium	Equilibrium constant	Assumed value at 5°C
(1) $\text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{OH}^-$	$K_w = [\text{H}^+][\text{OH}^-]$	$1.82 \times 10^{-15} \text{ mol}^2/\text{L}^2$
(2) $\text{SO}_2(\text{g}) + \text{H}_2\text{O} \leftrightarrow \text{SO}_2 \cdot \text{H}_2\text{O}$	$K_{\text{HS}} = \frac{P_{\text{SO}_2}}{[\text{SO}_2 \cdot \text{H}_2\text{O}]}$	0.379 atm L/mol
(3) $\text{SO}_2 \cdot \text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{HSO}_3^-$	$K_{\text{IS}} = \frac{[\text{H}^+][\text{HSO}_3^-]}{[\text{SO}_2 \cdot \text{H}_2\text{O}]}$	0.0206 mol/L
(4) $\text{HSO}_3^- \leftrightarrow \text{H}^+ + \text{SO}_3^{=}$	$K_{\text{2S}} = \frac{[\text{H}^+][\text{SO}_3^{=}]}{[\text{HSO}_3^-]}$	$8.88 \times 10^{-8} / \text{L}$
(5) $\text{NH}_3(\text{g}) + \text{H}_2\text{O} \leftrightarrow \text{NH}_3 \cdot \text{H}_2\text{O}$	$K_{\text{HN}} = \frac{P_{\text{NH}_3}}{[\text{NH}_3 \cdot \text{H}_2\text{O}]}$	$7.11 \times 10^{-3} \text{ atm L/mol}$
(6) $\text{NH}_3 \cdot \text{H}_2\text{O} \leftrightarrow \text{NH}_4^+ + \text{OH}^-$	$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3 \cdot \text{H}_2\text{O}]}$	$1.5 \times 10^{-5} \text{ mol/L}$
(7) $\text{CO}_2(\text{g}) + \text{H}_2\text{O} \leftrightarrow \text{CO}_2 \cdot \text{H}_2\text{O}$	$K_{\text{HC}} = \frac{P_{\text{CO}_2}}{[\text{CO}_2 \cdot \text{H}_2\text{O}]}$	16.6 atm L/mol
(8) $\text{CO}_2 \cdot \text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{HCO}_3^-$	$K_{\text{1C}} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2 \cdot \text{H}_2\text{O}]}$	$2.94 \times 10^{-7} \text{ mol/L}$
(9) $\text{HCO}_3^- \leftrightarrow \text{H}^+ + \text{CO}_3^{=}$	$K_{\text{2C}} = \frac{[\text{H}^+][\text{CO}_3^{=}]}{[\text{HCO}_3^-]}$	$2.74 \times 10^{-11} \text{ mol/L}$
(10) $\Sigma[\text{CO}_2] = C_1 (\text{mol}/\text{m}^3)$		
(11) $\Sigma[\text{S(IV)}] = C_2 (\text{mol}/\text{m}^3)$		
(12) $\Sigma[\text{N}(-\text{III})] = C_3 (\text{mol}/\text{m}^3)$		
(13) $\Sigma[\text{Strong acid anions}] = C_4 (\text{mol}/\text{m}^3)$		
(14) Liquid water content = (L/m^3)		
(15) $\Sigma[+] = \Sigma[-]$		

Now let us throw some sea salt containing CaCO_3 and MgCO_3 into the mixture. This provides alkalinity



The SO_4^{2-} is missing in action because it reacts with CaCO_3 and becomes $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (gypsum) a poorly soluble salt

The stronger acid pushes the weaker one out of its salt.
(also works with NaCl , in which case gas phase Cl_2 escapes)

FIG. 16-2 Same as [Fig. 16-1](#), except for the ca. 0.001 g/m^3 of water nucleated on alkaline seasalt particles in the same cloud. $1 \mu\text{g/m}^3$ of seasalt.

Now let us look at a polluted site and at acid rain

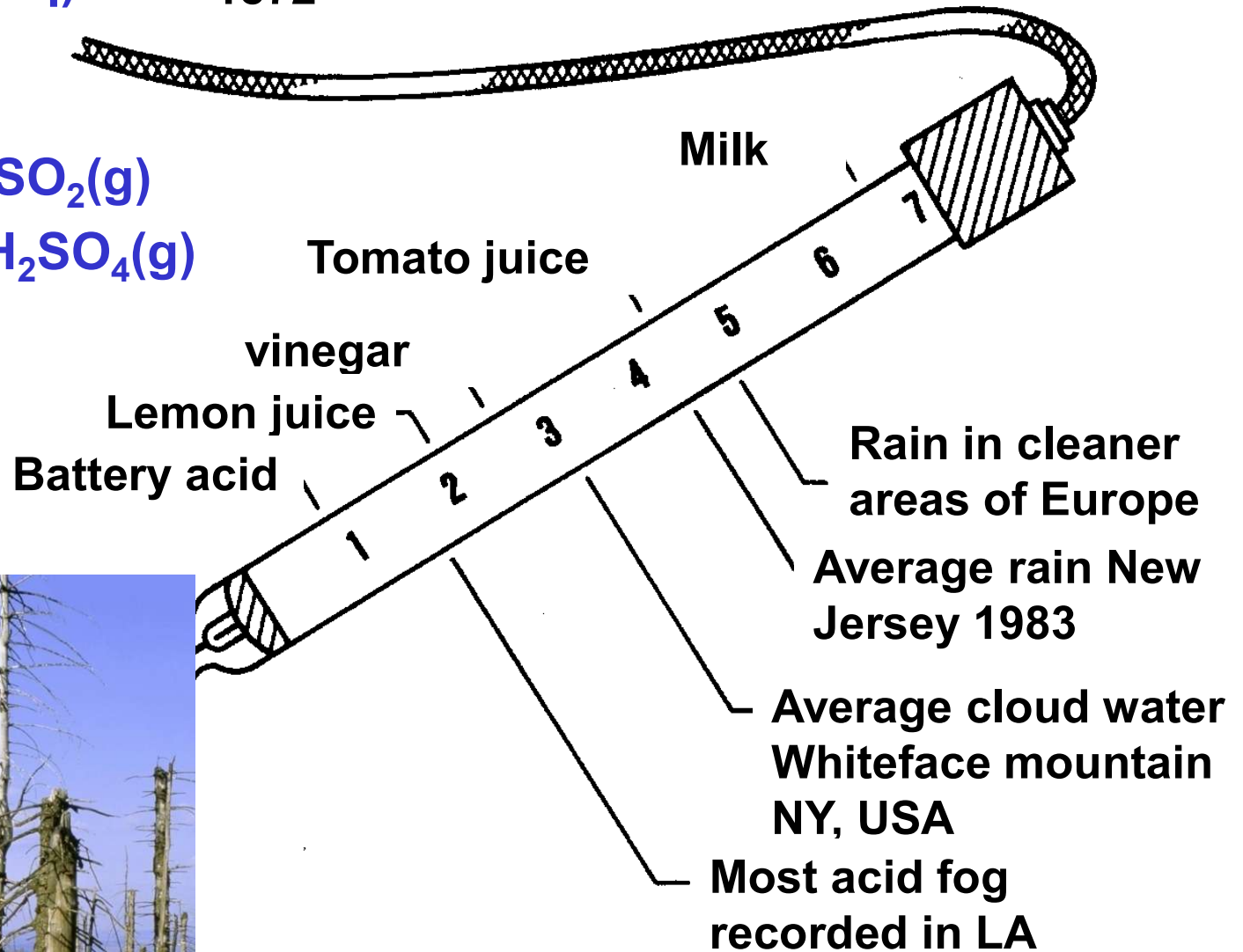
Dissolution (Henry's law)



Term "Acid Rain": Robert Angus Smith, 1872

In the gas phase gets $\text{SO}_2(\text{g})$

Gets oxidized to form $\text{H}_2\text{SO}_4(\text{g})$



How to calculate the pH of acid rain – step 1

Dissolution (Henry's law)



The equilibrium constant is expressed as Henry's law constant K_H which is equal to 1.0M/at at 25°C and is defined as

$$K_H = [\text{H}_2\text{SO}_3(\text{aq})]/P \text{ where } P \text{ is the atmospheric partial pressure of } \text{SO}_2$$

If you have to calculate the pH for an partial pressure of 100 ppb (Power plant plume) which is the same as $1.0 \times 10^{-7} \text{ atm}$ then

$$1.0 \times 10^{-7} \text{ M} = [\text{H}_2\text{SO}_3(\text{aq})] = PK_H$$

How to calculate the pH – step 2

Next you need the K_a of H_2SO_3 that is 1.7×10^{-2} then



Because of the equilibrium, between $\text{SO}_2(\text{g})$ and $\text{H}_2\text{SO}_3(\text{aq})$ the $[\text{H}_2\text{SO}_3(\text{aq})]$ corresponds to only the $\text{H}_2\text{SO}_3(\text{aq})$ that does not ionize to bisulfite and it remains constant at $1.0 \times 10^{-7} \text{M}$ as long as the gas phase SO_2 partial pressure does not change

$$K_a = 1.7 \times 10^{-2} = [\text{HSO}_3^-][\text{H}^+] / [\text{H}_2\text{SO}_3(\text{aq})] = [\text{HSO}_3^-]^2 / 1.0 \times 10^{-7}$$

$$\text{pH} = -\log_{10}(a_{\text{H}^+}) \approx -\log_{10}([\text{H}^+] / \text{M})$$

$$[\text{HSO}_3^-] = [\text{H}^+] = 4.1 \times 10^{-5} \text{ M}$$

$$\text{pH} = 4.38$$

Now let us look at a polluted site and at acid rain

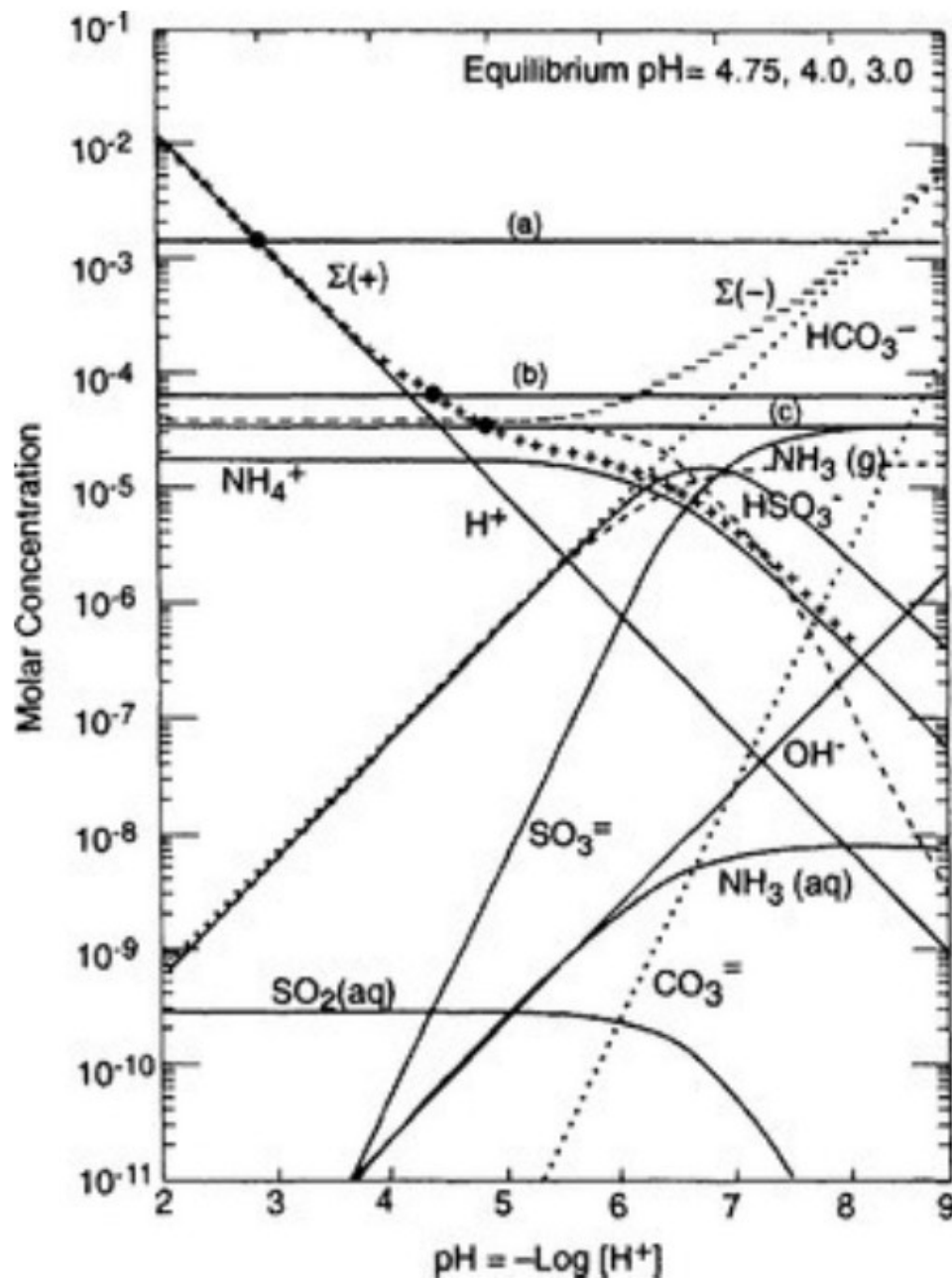
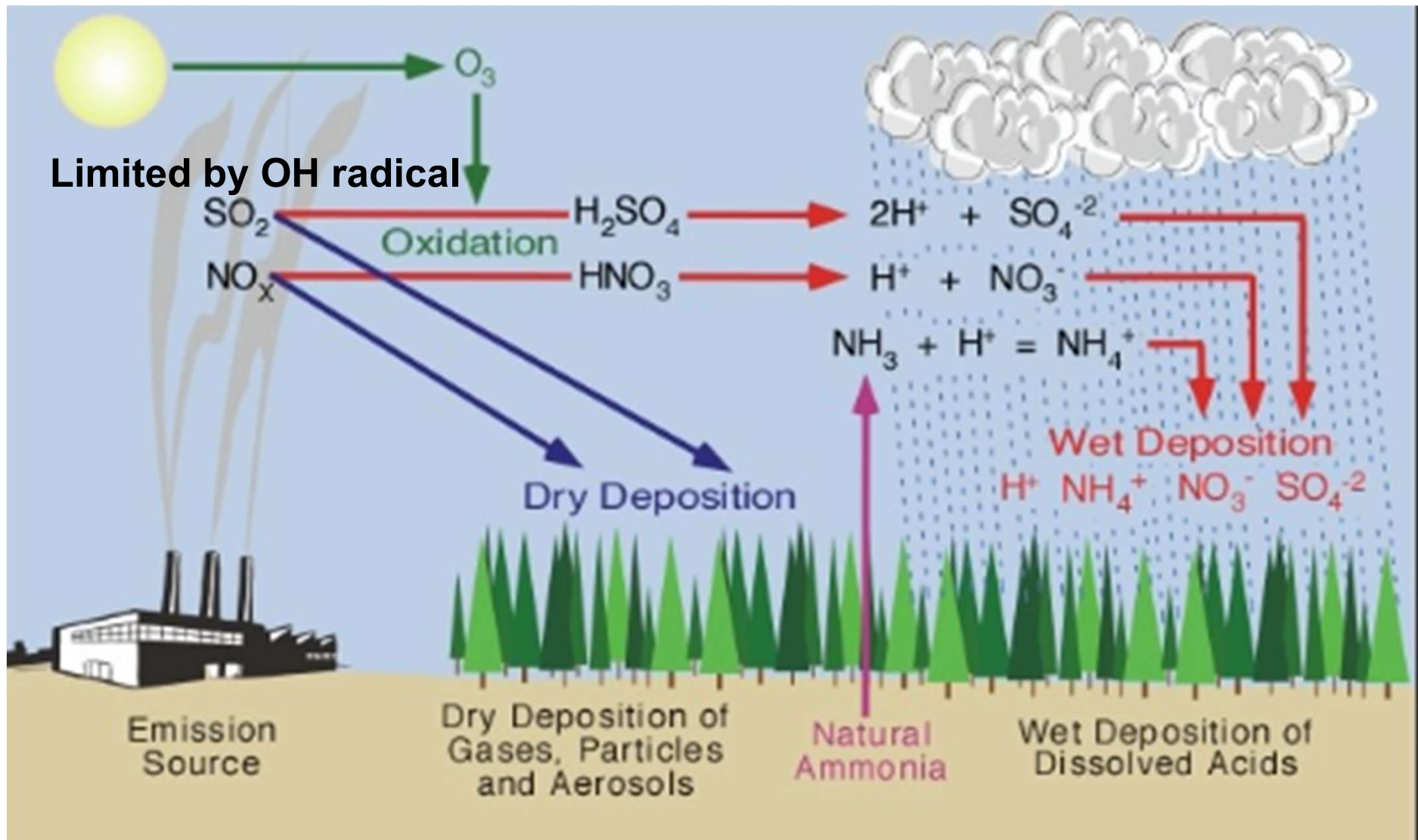


FIG. 16-3 Same as [Fig. 16-1](#), except (a) 10 $\mu\text{g}/\text{m}^3$ of H_2SO_4 aerosol; (b) 0.4 $\mu\text{g}/\text{m}^3$ and (c) 0.2 $\mu\text{g}/\text{m}^3$.

Through-fall (rain that has fallen through a canopy) is more acidic than rain in a clearing because SO_2 (g) and NO_2 (g) can, in the presence of H_2O (g), be oxidized by many oxidants including O_3 (g) on the leaf surface (multi-phase chemistry)



In river systems

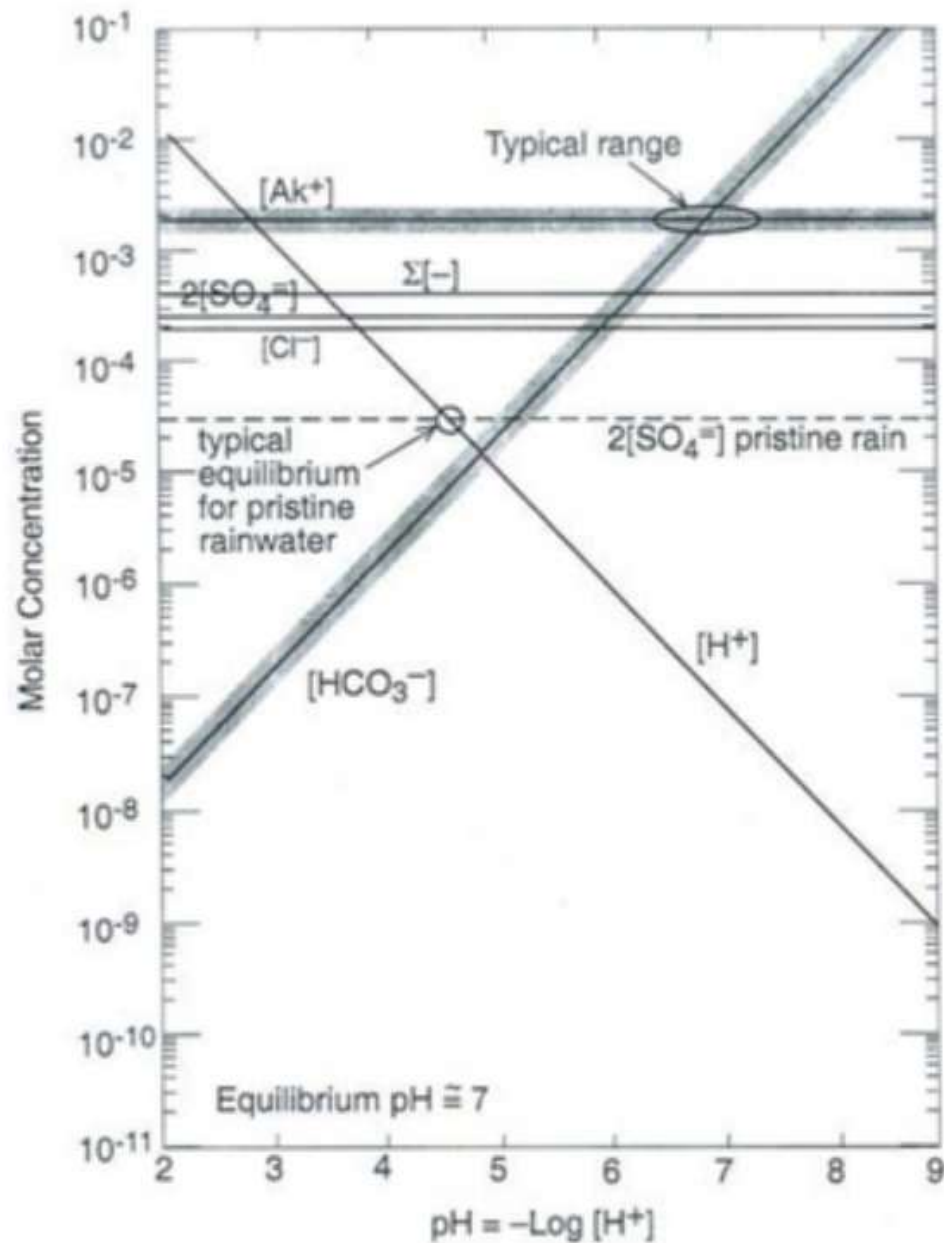


FIG. 16-5 Sillén (master variable) diagram for “average” river water, using data from [Stumm and Morgan \(1970\)](#).

In the ocean

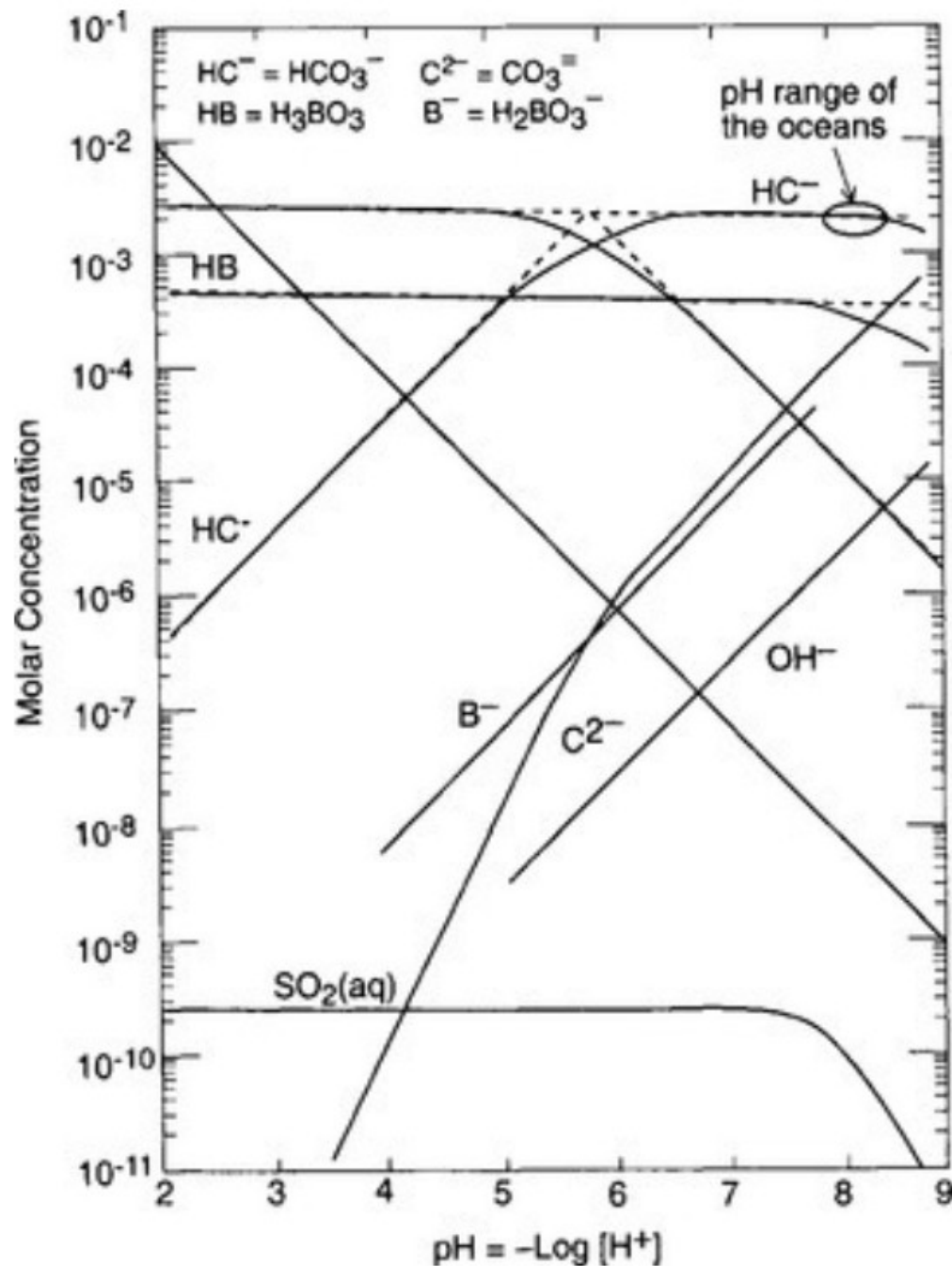


FIG. 16-6 Sillén (master variable) diagram for seawater. Replotted from Sillén's own graph ([Sillén, 1967](#)).

Oxidation states of key elements

Table 16-3

Oxidation states of key elements. The *oxidation state* of an element is its charge or valence state in various chemical species. Many elements have more than one oxidation state. In nature you can usually *determine the oxidation state* of an element in a compound by assuming O(–II) and H(I)

		Chemical species
Carbon	C(–IV)	CH ₄ (methane)
	C(0)	C (graphite), CH ₂ O (glucose or other carbohydrate)
	C(+IV)	CO ₂ , HCO ₃ [–] , CO ₃ ^{2–} (carbonate system)
Hydrogen	H(0)	H ₂ (hydrogen gas)
	H(I)	H ⁺ (hydrogen ion)
Nitrogen	N(–III)	NH ₃ , NH ₄ ⁺ (ammonia)
	N(0)	N ₂ (nitrogen gas)
	N(+V)	NO ₃ [–] (nitrate ion)
Oxygen	O(–II)	H ₂ O, CO ₂ , SO ₄ ^{2–}
	O(0)	O ₂ (oxygen gas)
Sulfur	S(–II)	H ₂ S, HS [–] (hydrogen sulfide)
	S(0)	S (elemental sulfur)
	S(+II)	S ₂ O ₃ ^{2–} (thiosulfate ion)
	S(+VI)	SO ₄ ^{2–} (sulfate ion)
Iron	Fe(II)	Fe ²⁺ , Fe(OH) ₂ , FeCO ₃ (s) (ferrous ion, ferrous hydroxide, ferrous carbonate solid)
	Fe(III)	Fe ³⁺ , Fe ₂ O ₃ (s) (ferric ion, ferric oxide solid)
Manganese	Mn(II)	Mn ²⁺ , MnCO ₃ (s) (manganous ion, manganous carbonate solid)
	Mn(IV)	MnO ₂ (s) (manganese dioxide solid)

Some important redox reactions

Table 16-4

Examples of redox reactions, consisting of an oxidation and reduction half reaction

(a) <i>Photosynthetic production of reduced carbon in carbohydrate</i>		
Carbon reduced	($\text{CO}_2 \rightarrow \text{C}$)	$\text{CO}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons \text{H}_2\text{O} + \text{CH}_2\text{O}$
Oxygen in water oxidized	($\text{H}_2\text{O} \rightarrow \text{O}_2$)	$2\text{H}_2\text{O} \rightleftharpoons \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$
Net		$\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{O}_2 + \text{CH}_2\text{O}$
(b) <i>Organic matter oxidation</i>		
Carbon oxidized	($\text{C} \rightarrow \text{CO}_2$)	$\text{CH}_2\text{O} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + 4\text{H}^+ + 4\text{e}^-$
Oxygen reduced	($\text{O}_2 \rightarrow \text{H}_2\text{O}$)	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$
Net		$\text{CH}_2\text{O} + \text{O}_2 \rightleftharpoons \text{CO}_2 + \text{H}_2\text{O}$
(c) <i>Iron oxidation</i>		
Iron oxidized	($\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$)	$4\text{Fe}^{2+} \rightleftharpoons 4\text{Fe}^{3+} + 4\text{e}^-$
Oxygen reduced	($\text{O}_2 \rightarrow \text{H}_2\text{O}$)	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$
Net		$4\text{Fe}^{2+} + \text{O}_2 + 4\text{H}^+ \rightleftharpoons 4\text{Fe}^{3+} + 2\text{H}_2\text{O}$
(d) <i>Sulfide oxidation</i>		
Sulfide oxidized	($\text{HS}^- \rightarrow \text{SO}_4^{2-}$)	$\text{HS}^- + 4\text{H}_2\text{O} \rightleftharpoons \text{SO}_4^{2-} + 9\text{H}^+ + 8\text{e}^-$
Oxygen reduced	($\text{O}_2 \rightarrow \text{H}_2\text{O}$)	$2\text{O}_2 + 8\text{H}^+ + 8\text{e}^- \rightleftharpoons 4\text{H}_2\text{O}$
Net		$2\text{O}_2 + \text{HS}^- \rightleftharpoons \text{SO}_4^{2-} + \text{H}^+$
(e) <i>Sulfate reduction</i>		
Carbon oxidized	($\text{C} \rightarrow \text{CO}_2$)	$2\text{CH}_2\text{O} + 2\text{H}_2\text{O} \rightleftharpoons 2\text{CO}_2 + 8\text{H}^+ + 8\text{e}^-$
Sulfate reduced	($\text{SO}_4^{2-} \rightarrow \text{H}_2\text{S}$)	$\text{SO}_4^{2-} + 10\text{H}^+ + 8\text{e}^- \rightleftharpoons \text{H}_2\text{S} + 4\text{H}_2\text{O}$
Net		$2\text{CH}_2\text{O} + \text{SO}_4^{2-} + 2\text{H}^+ \rightleftharpoons \text{H}_2\text{S} + 2\text{CO}_2 + 2\text{H}_2\text{O}$
(f) <i>Iron reduction</i>		
Carbon oxidized	($\text{C} \rightarrow \text{CO}_2$)	$\text{CH}_2\text{O} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + 4\text{H}^+ + 4\text{e}^-$
Iron reduced	($\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$)	$4\text{Fe}^{3+} + 4\text{e}^- \rightleftharpoons 4\text{Fe}^{2+}$
Net		$\text{CH}_2\text{O} + \text{H}_2\text{O} + 4\text{Fe}^{3+} \rightleftharpoons \text{CO}_2 + 4\text{H}^+ + 4\text{Fe}^{2+}$
(g) <i>Nitrate reduction</i>		
Carbon oxidized	($\text{C} \rightarrow \text{CO}_2$)	$5\text{CH}_2\text{O} + 5\text{H}_2\text{O} \rightleftharpoons 5\text{CO}_2 + 20\text{H}^+ + 20\text{e}^-$
Nitrate reduced	($\text{NO}_3^- \rightarrow \text{N}_2$)	$4\text{NO}_3^- + 24\text{H}^+ + 20\text{e}^- \rightleftharpoons 2\text{N}_2 + 12\text{H}_2\text{O}$
Net		$4\text{NO}_3^- + 5\text{CH}_2\text{O} + 4\text{H}^+ \rightleftharpoons 5\text{CO}_2 + 2\text{N}_2 + 7\text{H}_2\text{O}$

Gibbs free energy for some important redox reactions

Table I6-5

The standard free energy of reaction,

ΔG_r^0 , for the main environmental redox reactions

Reaction	ΔG_r^0 (1/2 reaction, kJ/mol)	ΔG^0 (whole reaction, kJ/mol)
<i>Oxidation</i>		
$\text{CH}_2\text{O}^a + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 4\text{H}^+ + 4\text{e}^-$	-27.4	
<i>Reduction</i> ^b		
$4\text{e}^- + 4\text{H}^+ + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$	-491.0	-518.4
$4\text{e}^- + 4.8\text{H}^+ + 0.8\text{NO}_3^- \rightarrow 0.4\text{N}_2 + 2.4\text{H}_2\text{O}$	-480.2	-507.6
$4\text{e}^- + 8\text{H}^+ + 2\text{MnO (s)} \rightarrow 2\text{Mn}^{2+} + 4\text{H}_2\text{O}$	-474.5	-501.9
$4\text{e}^- + 12\text{H}^+ + 2\text{Fe}_2\text{O}_3 \text{ (s)} \rightarrow 4\text{Fe}^{2+} + 6\text{H}_2\text{O}$	-253.22	-280.6
$4\text{e}^- + 5\text{H}^+ + \frac{1}{2}\text{SO}_4^{2-} \rightarrow \frac{1}{2}\text{H}_2\text{S} + 2\text{H}_2\text{O}$	-116.0	
$4\text{e}^- + 4\text{H}^+ + \text{CH}_2\text{O} \rightarrow \text{CH}_4 + \text{H}_2\text{O}$	-7.0	

^aCH₂O represents organic matter. $\Delta G_f = -129$ kJ/mol.

^bThe free energies of formation are (kJ/mol) ([Stumm and Morgan, 1981](#)): H⁺ = 0; MnO₂ (pyrolusite) = -465.1; Mn²⁺ = -228.0; H₂O = -237.0; NO₃⁻ = -111.3; N₂ = 0; Fe²⁺ = -73.9; Fe₂O₃

Reaction sequence in sediment systems

Reactions occur in the sequence in which the gain of free energy is the greatest

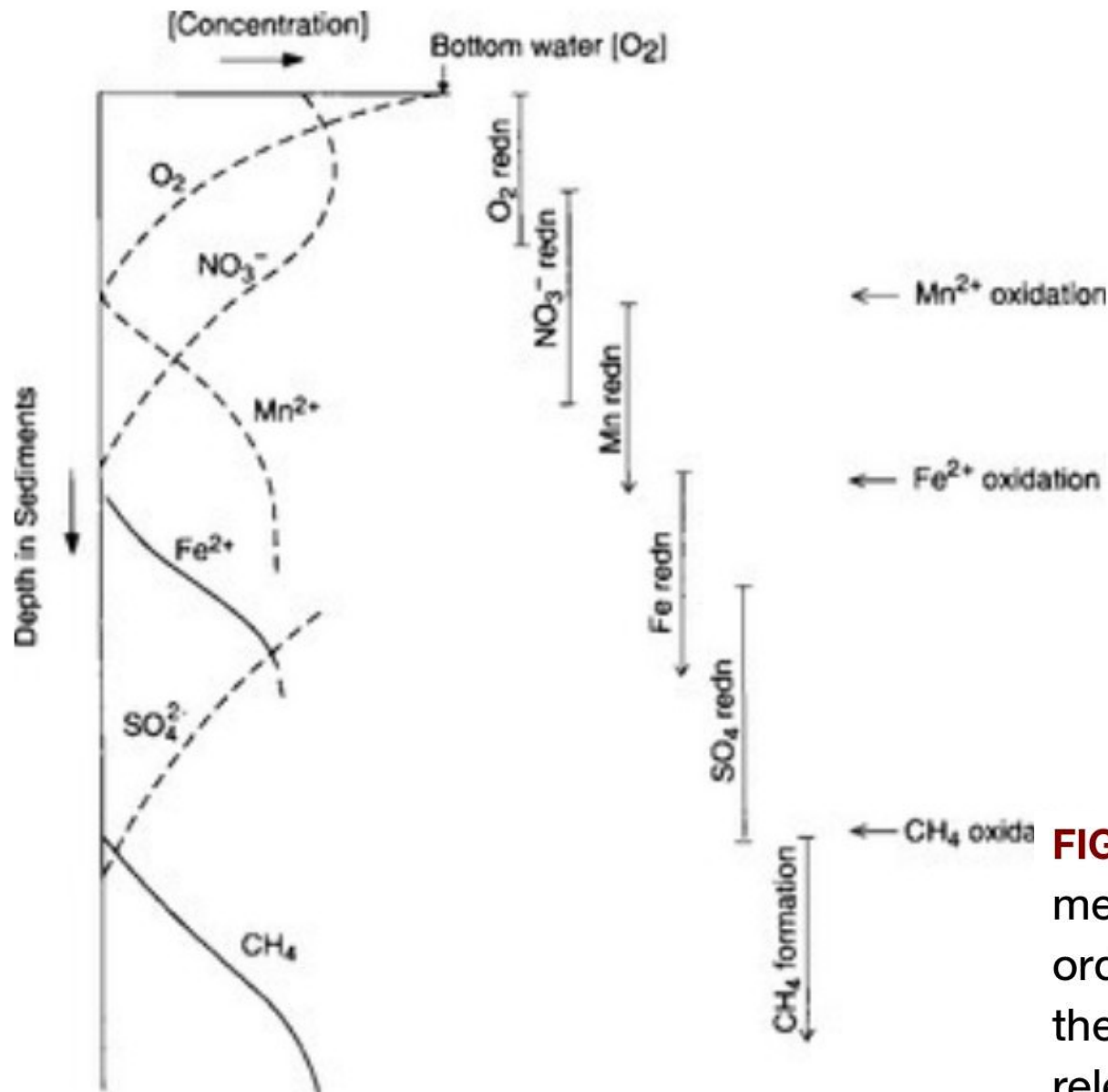


FIG. 16-7 Changes in concentration of metabolites in sediment. The reaction order follows the sequence dictated by thermodynamics (i.e., the most energy releasing reactions occur first). (Modified

from [Froelich et al., 1979](#) .)

Reaction sequence in sediment systems

Equation	$p\varepsilon^{0a}$	$p\varepsilon_w^{0b}$	$p\varepsilon_c^c$
$\frac{1}{4}\text{O}_2(\text{g}) + \text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2\text{O}$	+20.75	+13.35	+12.8
$\frac{1}{5}\text{NO}_3^- + \frac{6}{5}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{10}\text{N}_2(\text{g}) + \frac{3}{5}\text{H}_2\text{O}$	+21.05	+12.17	+11.2
$\frac{1}{6}\text{IO}_3^- + \text{H}^+ + \text{e}^- \rightarrow \frac{1}{6}\text{I}^- + \frac{1}{2}\text{H}_2\text{O}$	+18.35	+10.95	+10.5
$\frac{1}{2}\text{MnO}_2(\text{s}) + 2\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{Mn}^{2+} + \text{H}_2\text{O}$	+21.85	+7.05	+8.9
$\frac{1}{3}\text{CrO}_4^{2-} + 2\text{H}^+ + \text{e}^- \rightarrow \frac{1}{3}\text{Cr}(\text{OH})_2^+ + \frac{2}{3}\text{H}_2\text{O}$	+22.03	+7.23	+6.6
$\frac{1}{6}\text{N}_2(\text{g}) + \frac{4}{3}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{3}\text{NH}_4^+$	+4.65	-5.22	-2.6
$\text{FeOOH}(\text{s}) + 3\text{H}^+ + \text{e}^- \rightarrow \text{Fe}^{2+} + 2\text{H}_2\text{O}$	+8.25	-6.55	-2.7
$\frac{1}{8}\text{SO}_4^{2-} + \frac{9}{8}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{8}\text{HS}^- + \frac{1}{2}\text{H}_2\text{O}$	+4.25	-4.08	-3.4

Table 16-6

Redox half reactions, $p\varepsilon^0$, ε_w^0 , and calculated $p\varepsilon$ values from the redox species distributions $p\varepsilon_c$ for the zone 125–135 m in Saanich Inlet

$^a p\varepsilon^0 \equiv \log K$; $\log K$ estimated from free energy (ΔG_r^0) data. $p\varepsilon^0$ is the $p\varepsilon$ to expect at equilibrium if the ratio of the oxidized and reduced species are at unit activity:

Reaction sequence in sediment systems

Table 16-6

Redox half reactions, $p\epsilon^0$, ϵ_w^0 , and calculated $p\epsilon$ values from the redox species distributions $p\epsilon_c$ for the zone 125–135 m in Saanich Inlet

Equation	$p\epsilon^{0a}$	$p\epsilon_w^{0b}$	$p\epsilon_c^c$
$\frac{1}{4}\text{O}_2(\text{g}) + \text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2\text{O}$	+20.75	+13.35	+12.8
$\frac{1}{5}\text{NO}_3^- + \frac{6}{5}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{10}\text{N}_2(\text{g}) + \frac{3}{5}\text{H}_2\text{O}$	+21.05	+12.17	+11.2
$\frac{1}{6}\text{IO}_3^- + \text{H}^+ + \text{e}^- \rightarrow \frac{1}{6}\text{I}^- + \frac{1}{2}\text{H}_2\text{O}$	+18.35	+10.95	+10.5
$\frac{1}{2}\text{MnO}_2(\text{s}) + 2\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{Mn}^{2+} + \text{H}_2\text{O}$	+21.85	+7.05	+8.9
$\frac{1}{3}\text{CrO}_4^{2-} + 2\text{H}^+ + \text{e}^- \rightarrow \frac{1}{3}\text{Cr}(\text{OH})_3 + \frac{2}{3}\text{H}_2\text{O}$	+22.03	+7.23	+6.6
$\frac{1}{6}\text{N}_2(\text{g}) + \frac{4}{3}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{3}\text{NH}_4^+$	+4.65	-5.22	-2.6
$\text{FeOOH}(\text{s}) + 3\text{H}^+ + \text{e}^- \rightarrow \text{Fe}^{2+} + 2\text{H}_2\text{O}$	+8.25	-6.55	-2.7
$\frac{1}{8}\text{SO}_4^{2-} + \frac{9}{8}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{8}\text{HS}^- + \frac{1}{2}\text{H}_2\text{O}$	+4.25	-4.08	-3.4

^a $p\epsilon^0 \equiv \log K$; $\log K$ estimated from free energy (ΔG_r^0) data. $p\epsilon^0$ is the $p\epsilon$ to expect at equilibrium if the ratio of the oxidized and reduced species are at unit activity:

$$\ln \left\{ \frac{a_{\text{red}}}{a_{\text{ox}}} \right\} = \ln 1 = 0$$

^b $p\epsilon_w^0$ is the $p\epsilon^0$ normalized to constant pH: $\epsilon_w^0 = p\epsilon^0 + n(-7.4)$; pH here = 7.4.

^c $p\epsilon_c$ is the calculated $p\epsilon$ using the measured activities of the oxidized and reduced species:

$$p\epsilon_e = p\epsilon^0 - \log \left\{ \frac{a_{\text{red}}}{a_{\text{ox}} a_{\text{H}^+}^n} \right\}$$

assuming: pressure (N_2) = 0.76,
 $[\text{SO}_4^{2-}] = 28 \times 10^{-3} \text{ mol/kg}$; pH = 7.4.

Reactions in oxygen depleted waters

Table 16-6

Redox half reactions, $p\epsilon^0$, ϵ_w^0 , and calculated $p\epsilon$ values from the redox species distributions $p\epsilon_c$ for the zone 125–135 m in Saanich Inlet

Equation	$p\epsilon^{0a}$	$p\epsilon_w^{0b}$	$p\epsilon_c^c$
$\frac{1}{4}\text{O}_2(\text{g}) + \text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2\text{O}$	+20.75	+13.35	+12.8
$\frac{1}{5}\text{NO}_3^- + \frac{6}{5}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{10}\text{N}_2(\text{g}) + \frac{3}{5}\text{H}_2\text{O}$	+21.05	+12.17	+11.2
$\frac{1}{6}\text{IO}_3^- + \text{H}^+ + \text{e}^- \rightarrow \frac{1}{6}\text{I}^- + \frac{1}{2}\text{H}_2\text{O}$	+18.35	+10.95	+10.5
$\frac{1}{2}\text{MnO}_2(\text{s}) + 2\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{Mn}^{2+} + \text{H}_2\text{O}$	+21.85	+7.05	+8.9
$\frac{1}{3}\text{CrO}_4^{2-} + 2\text{H}^+ + \text{e}^- \rightarrow \frac{1}{3}\text{Cr}(\text{OH})_3 + \frac{2}{3}\text{H}_2\text{O}$	+22.03	+7.23	+6.6
$\frac{1}{6}\text{N}_2(\text{g}) + \frac{4}{3}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{3}\text{NH}_4^+$	+4.65	-5.22	-2.6
$\text{FeOOH}(\text{s}) + 3\text{H}^+ + \text{e}^- \rightarrow \text{Fe}^{2+} + 2\text{H}_2\text{O}$	+8.25	-6.55	-2.7
$\frac{1}{8}\text{SO}_4^{2-} + \frac{9}{8}\text{H}^+ + \text{e}^- \rightarrow \frac{1}{8}\text{HS}^- + \frac{1}{2}\text{H}_2\text{O}$	+4.25	-4.08	-3.4

^a $p\epsilon^0 \equiv \log K$; $\log K$ estimated from free energy (ΔG_r^0) data. $p\epsilon^0$ is the $p\epsilon$ to expect at equilibrium if the ratio of the oxidized and reduced species are at unit activity:

$$\ln \left\{ \frac{a_{\text{red}}}{a_{\text{ox}}} \right\} = \ln 1 = 0$$

^b $p\epsilon_w^0$ is the $p\epsilon^0$ normalized to constant pH: $\epsilon_w^0 = p\epsilon^0 + n(-7.4)$; pH here = 7.4.

^c $p\epsilon_c$ is the calculated $p\epsilon$ using the measured activities of the oxidized and reduced species:

$$p\epsilon_e = p\epsilon^0 - \log \left\{ \frac{a_{\text{red}}}{a_{\text{ox}} a_{\text{H}^+}^n} \right\}$$

assuming: pressure (N_2) = 0.76,
 $[\text{SO}_4^{2-}] = 28 \times 10^{-3} \text{ mol/kg}$; pH = 7.4.

Depth distribution of species

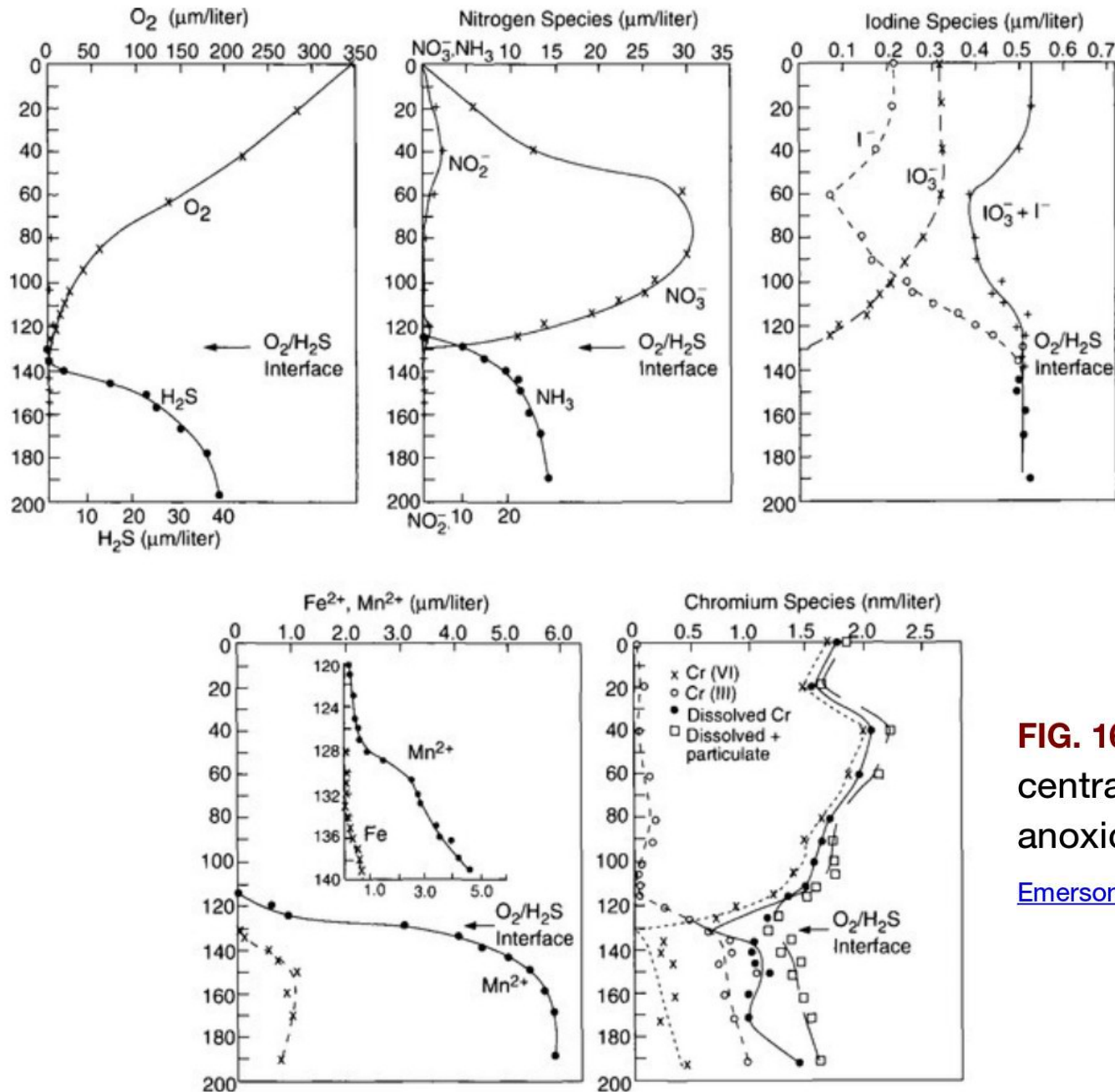


FIG. 16-8 Depth distribution of the concentration of redox couples in a partially anoxic fjord, Saanich Inlet. (Redrawn from [Emerson et al., 1979](#).)

The global oxygen cycle

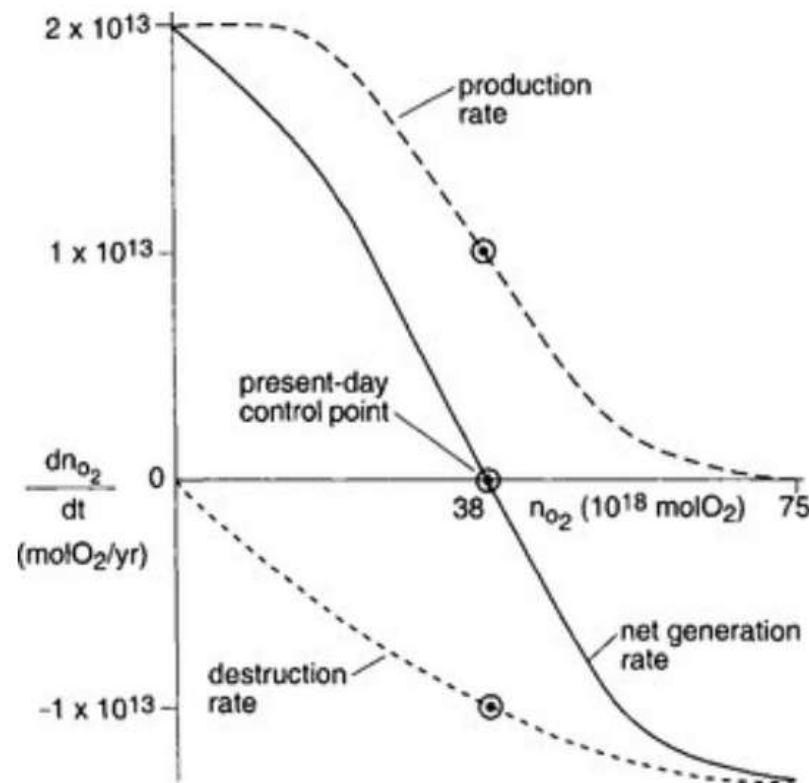


FIG. 16-10 A schematic representation of how the oxygen reduction and production rate, dn_{O_2}/dt combine to determine the size of the global oxygen reservoir. (Adapted from [Holland, 1978](#) .)

The global natural oxygen cycle

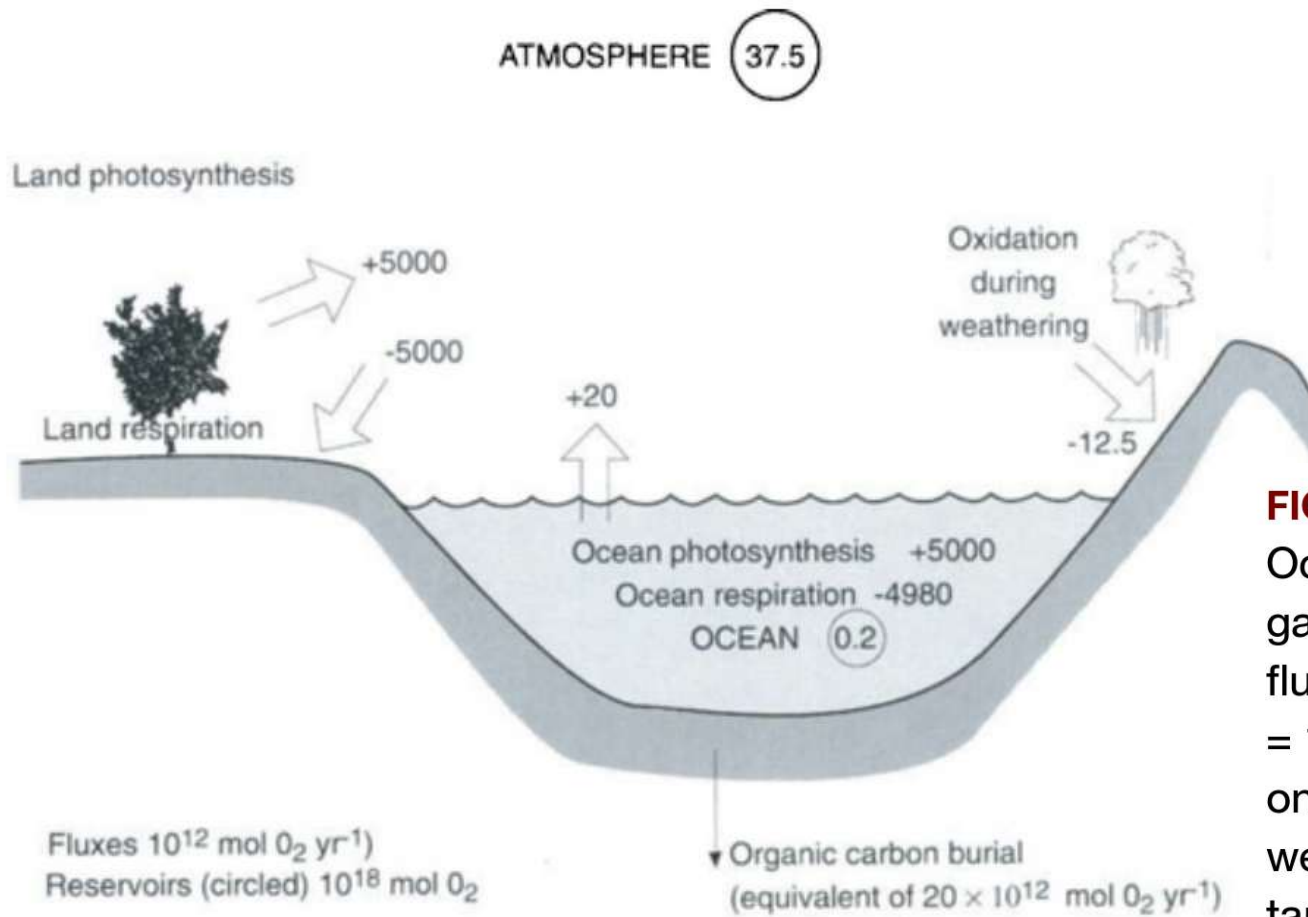


FIG. 16-9 The global oxygen balance. Ocean photosynthesis is estimated. Organic C fluxes are converted to oxygen fluxes using a stoichiometry of $\Delta\text{O}_2/\Delta\text{OC} = 1.5$ in the marine environment and 1.1 on land. Oxygen consumption during weathering of C, S, and Fe in sedimentary rocks, in the ratio of 6:2:1, is. At steady state organic carbon burial and O_2 consumption during weathering must be equal, which is certainly possible within the errors of these two estimates.

from satellite color ([Antoine et al., 1996](#)). It is assumed that photosynthesis on land and in the ocean are about equal. The global carbon burial rate is from [Hedges and Keil \(1995\)](#) from [Holland \(1978\)](#)

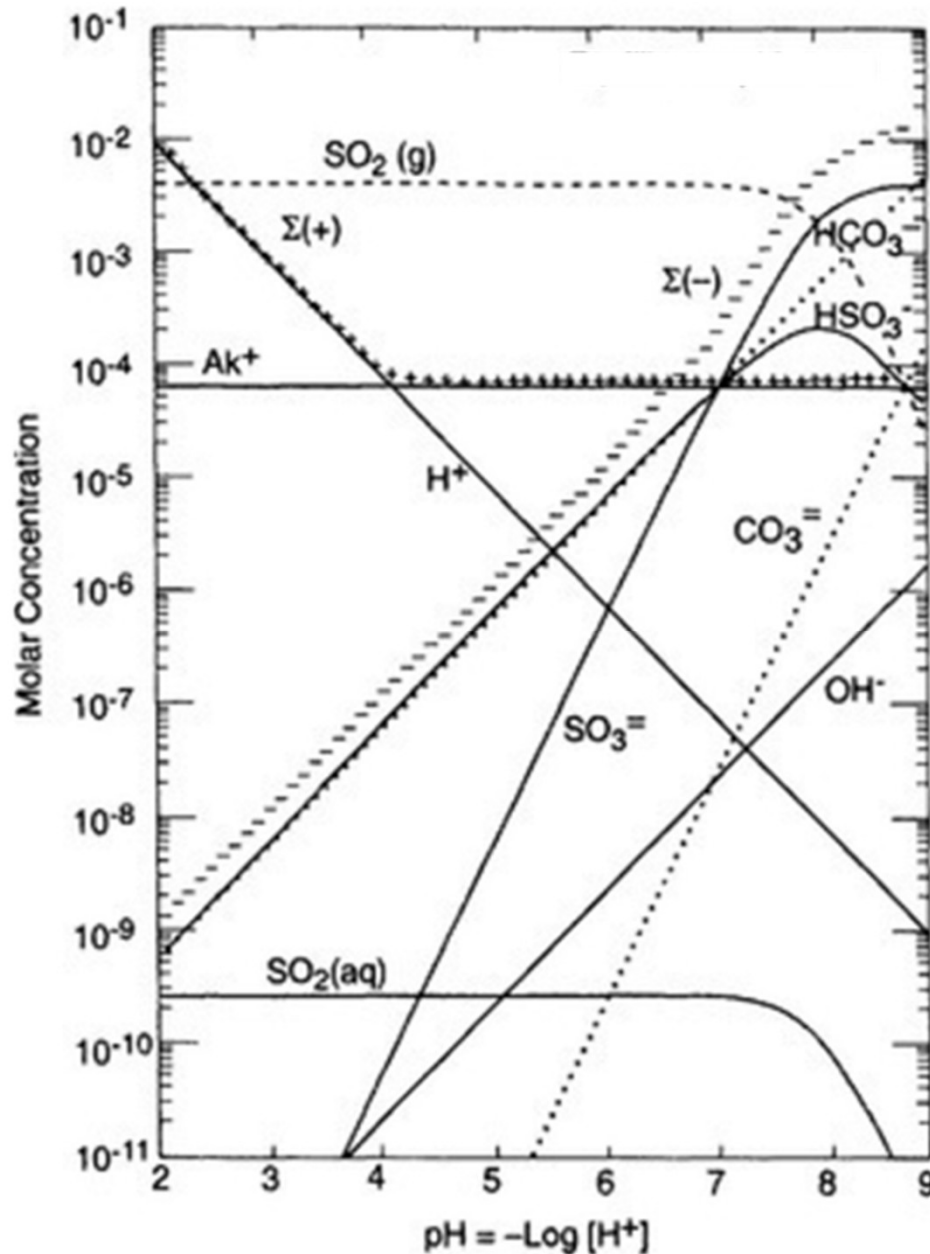
PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 15th September if you want to get them corrected before the Midsem exam

Numerical

1. If there is $5 \mu\text{g}/\text{m}^3$ of ammonium sulfate $(\text{NH}_4)_2\text{SO}_4$ aerosol in the air entering a cloud with $1 \text{ mL}/\text{m}^3$ liquid water content and $\epsilon = 1$ then what will the molarity of the cloud water?
2. How many M of H_2CO_3^* will be there in rainwater in equilibrium with 420 ppm CO_2 (present day levels) and 1000 ppm of CO_2 (projected 2100 levels for business as usual). Also calculate the pH of rainwater containing only carbonic acid for both cases. $K_H = 29 \text{ atm mol}^{-1} \text{ L}$, $K_1 = 4.5 \times 10^{-7}$
3. Calculate the pH of rainwater that is in equilibrium with an air that has an SO_2 concentration of 50 ppbv. $K_H = 1.0 \text{ mol L}^{-1} \text{ atm}^{-1}$, $K_1 = 1.7 \times 10^{-2}$.
4. You have measured a pH of 4.2 in rainwater on your campus. Assuming the acidity is from SO_2 then what is the concentration of SO_2 that must be in the air in ppbv
5. Oxygen in air is 21%. Calculate the molar concentration of dissolved oxygen in water that is saturated. $K_H = 1.28 \times 10^{-3} \text{ mol L}^{-1} \text{ atm}^{-1}$

Short answer answer



1. Read the equilibrium pH from the Sillen diagram
2. What are the oxidation states of carbon?
3. What are the oxidation states of sulfur?
4. What are the oxidation states of nitrogen?
5. What are the oxidation states of hydrogen?
6. What are the oxidation states of oxygen?
7. What are the oxidation states of iron?
8. What are the oxidation states of manganese?

Long answer

1. Why is the oxygen source flux to the atmosphere at any point in geological time limited by the organic carbon burial rates?
2. Why did snowball earth events only occur from the late Archean to the Proterozoic era when autotrophic photosynthetic organisms outnumbered heterotrophic life forms. Why did snowball earth never repeat after the Cambrian explosion, i.e. after the evolution of multicellular heterotrophic life forms?
3. What would happen if a drop in ocean pH would reduce the abundance of autotrophic plankton species, most of which have a mineral exoskeleton made of CaCO_3 or opal? How would this alter the oxygen cycle?
4. Fossil fuel combustion currently consumes $\sim 10^{15}$ mol of O_2 per year. Is this flux significant? Is the O_2 concentration in the atmosphere currently stable or not? What is the lifetime of in the atmospheric O_2 with respect to this flux and what would happen if fossil fuel combustion continued at the present rates for a couple of millenia?